

Sophia N. Wilson

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Education

- 2025–present **PhD Fellow, Sustainable Artificial Intelligence, SAINTS Lab, Department of Computer Science, University of Copenhagen (UCPH), Denmark.**
Resource efficient machine learning (ML), data efficient ML, data lifecycle.
- 2023–2025 **Master in Computational Physics, Niels Bohr Institute, UCPH, Denmark.**
CGPA: 12/12. Physics-informed ML, applied ML, applied statistics, scientific programming, cosmology.
- 2019–2023 **Bachelor in Physics, Niels Bohr Institute, UCPH, Denmark.**
CGPA: 11/12. Computational physics, statistics, numerical methods, astrophysics, cosmology.

Publications

In Conference Proceedings

- 2026 **Sophia N. Wilson**, Sebastian Mair, Mophat Okinyi, Erik B Dam, Janin Koch, and Raghavendra Selvan. How Hyper-Datafication Impacts the Sustainability Costs in Frontier AI. In *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. ACM, June 2026.
- 2026 **Sophia N. Wilson**, Guðrún Fjóra Guðmundsdóttir, Andrew Millard, Raghavendra Selvan, and Sebastian Mair. Position: Stop preaching and start practising data frugality for responsible development of ai. In *Proceedings of the 43rd International Conference on Machine Learning (ICML), Position Paper Track*, volume 306 of *Proceedings of Machine Learning Research*, 2026.

Preprints

- 2025 **Sophia N. Wilson**, Jens Hesselbjerg Christensen, and Raghavendra Selvan. Trading carbon for physics: On the resource efficiency of machine learning for spatio-temporal forecasting, 2025.

Research Projects

Department of Computer Science, UCPH

- Fall 2025 **Costs of Data Abundance and Benefits of Data Frugality in AI.**

Led two studies: (1) quantified environmental, social, and economic costs of hyperdatafication through large-scale metadata analysis of 550,000 Hugging Face datasets and a survey examining working conditions among data workers in Kenya; (2) developed and presented recommendations promoting a data-frugal AI approach at the Rethinking AI Workshop at EurIPS 2025.

Niels Bohr Institute & Department of Computer Science, UCPH

- Fall, 2024 – Spring, 2025 **Quantifying the Reduction in Carbon Footprint using Physics-Informed ML.**
Completed my MSc thesis applying physics-informed ML to reduce energy consumption in PDE simulations.

International Experience

- Feb. 2026 Machine Learning Summer School hosted by **Maincode, Melbourne, Australia.**
- Spring 2024 Exchange semester at the **Institute of Physics, University of Amsterdam, The Netherlands.**
- Aug. 2022 Astromatic Machine Learning Summer School at the **University of Montreal, Canada.**

Aug. 2021 Nordic Optical Telescope Summer School **La Palma, Spain.**

Teaching Experience

- 2026 **Data Analysis and Machine Learning (for high school teachers)**, *Department of Computer Science, Aarhus University.*
- 2025 **Machine Learning A**, *Department of Computer Science, UCPH.*
- 2025 **Mathematics A**, *Falkonergårdens Gymnasium.*
- 2023 **Summer Science Camp**, *Insero.*
- 2022-23 **Mathematics for Physicists 2**, *Niels Bohr Institute, UCPH.*
- 2022-23 **Mathematics for Physicists 1**, *Niels Bohr Institute, UCPH.*
- 2020-21 **Mechanics and Special Relativity**, *Niels Bohr Institute, UCPH.*

Supervision

- 2026 **Co-supervising two BSc students**, *Faculty of Humanities, UCPH.*

Guest Lecturing, Workshops, and Outreach

- 2026 **Co-organiser and moderator at the SAINTS'26 workshop on Sustainable AI**, *SAINTS Lab, Department of Computer Science, UCPH.*
- 2026 **Guest-lecturing on AI, Resource Use and Sustainability**, *Københavns Professionshøjskole.*
- 2026 **Organiser of a workshop on Environmental Impacts of AI**, *Det Åbne Gymnasium.*
- 2025 **Guest-lecturing in PhD course on Machine Learning for Sciences**, *Department of Computer Science, UCPH.*
- 2025 **Talks about Studying Physics at UCPH**, *Falkonergårdens Gymnasium.*
- 2025 **Presentation of current MSc thesis work at the 4th Annual Niels Bohr Institute MSc Student Symposium**, *Niels Bohr Institute, UCPH.*
- 2021-24 **Member of Women in Physics contributing to initiatives promoting gender balance in STEM**, *Niels Bohr Institute, UCPH.*

Invited Talks and Panel Debates

- 2026 **Speaker at the P1 Workshop on Green AI**, *Pioneer Center for AI.*
- 2026 **Speaker and panellist at Sustainability Week**, *Lund University.*
- 2025 **Speaker at resource-efficient AI workshop**, *D3A conference.*
- 2025 **Speaker at Climate Action Day**, *Copenhagen Center for Social Data Science, UCPH.*

Referees

Raghavendra Selvan

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UCPH*

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Jens Hesselbjerg Christensen

*Professor
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